

Urban Flood Nowcasting System - Presentation Speech

Good morning/afternoon, everyone.

Imagine a typical afternoon in Delhi. The sky is clear, the streets are bustling, and people are heading home after work. Suddenly, heavy rainfall bursts down. Within minutes, water starts pooling on roads, low-lying areas become lakes, traffic snarls, and commuters are left stranded. In those moments, residents have no quick way to know which neighbourhoods are at risk, nor how the situation will evolve over the next few hours.

That is exactly where our solution steps in.

Hello, I am *[Your Name]*, and today I will present our entry for the Smart India Hackathon 2026 - Problem Statement 26085: Urban Flood Nowcasting System.

The challenge isn't simply to detect rain. It's to understand where flooding may occur and how it will change in the imminent 0-3-hour window. Right now, weather data, terrain maps, and flood-risk information exist in silos. Decision-makers and citizens must piece them together manually, which is time-consuming and error-prone. Our goal is to integrate these disparate sources into a single, real-time platform that delivers clear, actionable insights.

We have built an interactive, web-based Urban Flood Nowcasting System that:

1. Collects data from multiple feeds - real-time weather, radar imagery, digital elevation models, and local drainage networks.
2. Processes the data through a robust pipeline and two analytical models: a Random-Forest Flood-Risk Classifier that flags high-risk zones using DEM and historic rainfall, and a Nowcasting Model that predicts rainfall and flood conditions for the next three hours using historic patterns and SWMM outputs.
3. Visualises the outcomes on an intuitive dashboard, offering both a quick-look view for the general public and a detailed technical view for experts.

All of this runs on a lightweight stack: pure HTML, vanilla CSS, and JavaScript on the front-end, with a Node-Express back-end deployed on Google Cloud Run.

One of the core features is Live Weather Mode. We pull data from the OpenWeather API every few minutes and overlay it on a dynamic map built with Leaflet.js and Mapbox tiles. Users can instantly see where it's raining right now, which is the first clue for impending flooding.

Our system ingests several data streams: real-time weather observations; Doppler radar imagery; DEM data from ISRO Bhuvan and NASA SRTM; SWMM-driven drainage data. These feeds feed into two principal models: Random-Forest Flood-Risk Classifier learns from terrain elevation and past rainfall to highlight vulnerable zones; Weather & Rainfall Nowcasting Model forecasts the next three-hour window using historic rainfall trends and SWMM outputs.

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Frontend: HTML, vanilla CSS, and plain JavaScript - no heavy frameworks, ensuring fast load times and easy maintenance.

Backend: Node.js with Express, containerised and served via Google Cloud Run for seamless scaling.

Deployment: static front-end lives on Vercel; all data are stored in Google Cloud Storage and queried from BigQuery.

Testing: we run unit- and integration-tests with Jest for JavaScript and Pytest for any Python components.

API Documentation: fully described in Swagger UI for developers to consume our endpoints.

We overlay Doppler radar imagery directly on the map, visualising rainfall intensity in real time. The dashboard offers two distinct user experiences: Simple View - a clean, high-level snapshot of flood risk, perfect for the general public or first-responders who need quick situational awareness; Technical View - a deep-dive that exposes water-level readings, rainfall intensity graphs, SWMM flow data, DEM profiles, and nowcasting predictions for power users and engineers.

Our Swagger UI exposes four key REST endpoints: Current Weather - fetches live weather for any latitude/longitude; Radar Data - returns the latest radar image or raw data; Flood Nowcasting - provides predicted flood extents for the next 0-3 hours; Dashboard Summary - aggregates all relevant metrics into a concise JSON payload. When an endpoint is called, the back-end pulls the required data, runs the appropriate model, and returns a structured JSON response. Additionally, a chatbot is integrated into the UI to answer user queries, guide navigation, and provide on-the-fly explanations.

Our solution directly addresses three critical outcomes: minimise water contamination by early identification of flood-prone zones; improve public safety through timely alerts about water depth and flood progression; save lives by empowering citizens and authorities with actionable, up-to-the-minute information.

In summary, we fuse real-time weather monitoring, radar visualisation, terrain and drainage data, flood-risk analytics, and short-term nowcasting into a single, user-friendly platform. This provides faster awareness and stronger decision-support for both citizens and emergency managers during urban flooding events.

Thank you for your attention. I am happy to answer any questions.